Keywords: digital dentistry; Digital smile-design; digital workflow; Guided crown-lengthening; Guided Implantology; Multidisciplinary approach

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doi: 10.1016/j.jdent.2022.104038

Journal of Dentistry 121 (2022) 104039

Accuracy of CNN-Based method for segmentation of the mandible

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Aims: This study aimed to evaluate the accuracy of a new automatic deep learning-based approach for fully mandible automatic segmentation from CBCTs.
Methods: Forty CBCT scans from healthy patients (20 females, 20 males, mean age 23.37 ± 3.34 years) were collected and a manual mandible segmentation was carried out by using Mimics software. Twenty CBCT scans were randomly selected and used for training the artificial intelligence model file. The remaining 20 CBCT segmentation masks were used to test the accuracy of the CNN fully automatic method by comparing the segmentation volumes of the 3D models obtained with automatic and manual segmentations. The accuracy of this method was also assessed by using the DICE Score coefficient (DSC) and by the surface-to-surface matching technique. The Intraclass correlation coefficient (ICC) and Dahlberg's formula were used to test the intra-observer reliability and method error, respectively. Independent Student's t-test was used for between-groups volumetric comparison.
Results: Measurements were highly correlated with an ICC value of 0.937 while the method error was 0.24 mm^3. An insignificant statistical difference of 0.71 (± 0,49) cm^3 was found between the methodologies ($p>0.05$). The matching percentage detected was 90.35% ± 1.88 (tolerance, 0.5 mm) and 96.32% ± 1.97% (tolerance 1.0 mm). All volumetric measurements were highly correlated with an ICC value of 0.937 for intra-observer evaluation and 0.919 for inter-observer evaluation.
Conclusions: The tested technology provided an accurate fully automated segmentation of the mandible from CBCT scans performing at a much higher speed than an experienced image reader.

Keywords: convolutional; mandible; networks; neural; segmentation

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doi: 10.1016/j.jdent.2022.104039

Journal of Dentistry 121 (2022) 104040

Full Digital rehabilitation in TMD patient

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Purpose: The purpose of the current work is to verify the feasibility of a digital articulator assembly for the diagnosis and therapy of patients with TMD.
Methods: They were used for optical impression (Carestream intraoral scanner), Condylography (Cadiax 4 Diagnostic Gamma), facial scanning (Carestream CS 9600), mounting in an articulator (Reference SL), for CAD processing /CAM (Exocad), for recording functional movements (MODJAW).
Results: the digital recordings are reliable and stackable on the analog recordings, for the construction of prosthetic and therapeutic artifacts in accordance with the occlusal concepts of the Vienna school it was necessary to find skills and customizations in the management of the CAD/CAM software.
Conclusions: Digital technology helps and amplifies the diagnostic possibilities in both general and dysfunctional patients allowing the reproducibility of the acquired data and their optimization. There is still no standard protocol that can be brought into daily practice.

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Keywords:

doi: 10.1016/j.jdent.2022.104040

Journal of Dentistry 121 (2022) 104041

Rotational Positioning of One-piece Implants in Computer-Guided Flapless Surgery and Immediate Loading Using the All-on-4 Concept

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Purpose: The aim of this clinical application is to assess the procedure of simultaneous insertion and rotational positioning of the one-piece implant (implant + Multi-Unit-Abutment) in order to achieve the right emergence angle of the abutment using reference points on the surgical guide. This clinical procedure is particularly useful in cases of immediate loading using the All-on-4 concept.
Methods: The case is designed through dedicated software by virtually inserting the implants in the optimal position with respect to the prosthesis and the bone situation. The program automatically determines the position of the bushing and virtual modeling, printing and gluing of the bushing to the surgical guide are performed.
Results: The procedure ensures correct integrated Multi-Unit-Abutment rotational positioning done in one movement using both implant contra-angle or manual insertion. Results are appreciable both for the precision of the abutment emergence angles for prosthetic long-term biomechanical stability and for the absence of bacterial microfiltration in the abutment/implant interface. The clinical sequence is easy for the surgeon and time-saving.
Conclusions: The surgeon is less stressed-out and the time needed is much less than standard procedure with Multi-Unit-Abutments screwing. This is facilitated by the reference points on the bushing. At the same time, further scientific evidence is needed and also the procedure must be integrated with the drilling undersized protocol for achieving the proper primary stability for All-on-4 immediate loading purposes. We think that this application needs clinical attention for reducing chair-time and stress-related procedures.

Keywords: abutment emergence; One-piece implant; reference point; rotational positioning

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doi: 10.1016/j.jdent.2022.104041